

Polymer Chemistry

MCQ

1. A polymer is made up of
 - a) Single atom
 - b) Large molecules with repeating units
 - c) Ions only
 - d) Crystals

Answer: b) Large molecules with repeating units

2. The word polymer is derived from
 - a) Latin
 - b) English
 - c) Greek
 - d) French

Answer: c) Greek

3. The term “poly” means
 - a) One
 - b) Few
 - c) Many
 - d) Small

Answer: c) Many

4. The term “mer” refers to
 - a) Bond
 - b) Chain
 - c) Repeating unit
 - d) Electron

Answer: c) Repeating unit

5. Polymers are formed by the process of

- a) Condensation
- b) Oxidation
- c) Polymerization
- d) Neutralization

Answer: c) Polymerization

6. Monomers are joined together by

- a) Ionic bonds
- b) Metallic bonds
- c) Covalent bonds
- d) Hydrogen bonds

Answer: c) Covalent bonds

7. Covalent bond is formed by

- a) Transfer of electrons
- b) Sharing of electrons
- c) Loss of protons
- d) Gain of neutrons

Answer: b) Sharing of electrons

8. A polymer is best defined as

- a) A mixture
- b) A short chain compound
- c) A long chain compound
- d) An element

Answer: c) A long chain compound

9. Which of the following is a natural polymer

- a) Nylon

- b) PVC
- c) Cellulose
- d) Teflon

Answer: c) Cellulose

10. Which of the following is a synthetic polymer

- a) Protein
- b) DNA
- c) Chitin
- d) Nylon

Answer: d) Nylon

11. Natural polymers are found mainly in

- a) Rocks
- b) Living organisms
- c) Plastics
- d) Metals

Answer: b) Living organisms

12. Cellulose is mainly found in

- a) Wool
- b) Cotton
- c) Silk
- d) Plastic

Answer: b) Cotton

13. Chitin is found in

- a) Plant roots
- b) Human bones
- c) Insect exoskeleton

d) Plastic bags

Answer: c) Insect exoskeleton

14. Lignin provides

a) Colour to plants

b) Structure to plants

c) Food to plants

d) Water to plants

Answer: b) Structure to plants

15. Proteins are made up of

a) Fatty acids

b) Glucose

c) Amino acids

d) Nucleotides

Answer: c) Amino acids

16. Number of different amino acids forming proteins is

a) 10

b) 15

c) 20

d) 25

Answer: c) 20

17. DNA is an example of

a) Synthetic polymer

b) Protein polymer

c) Natural polymer

d) Plastic

Answer: c) Natural polymer

18.Synthetic polymers are prepared from

- a) Plants
- b) Animals
- c) Petroleum products
- d) Minerals

Answer: c) Petroleum products

19.Ethylene is used to make

- a) Glass
- b) Plastic bags
- c) Cement
- d) Paper

Answer: b) Plastic bags

20.PVC is mainly used for

- a) Cooking utensils
- b) Pipes and window frames
- c) Clothes
- d) Medicines

Answer: b) Pipes and window frames

21.Fibres are

- a) Thick and rigid
- b) Short structures
- c) Thread-like structures
- d) Powdery substances

Answer: c) Thread-like structures

22.Fibres can be spun into

- a) Sheets

- b) Yarn
- c) Blocks
- d) Pellets

Answer: b) Yarn

23. Fibres are classified based on

- a) Colour
- b) Size
- c) Origin
- d) Cost

Answer: c) Origin

24. Cotton is a

- a) Animal fibre
- b) Mineral fibre
- c) Plant fibre
- d) Synthetic fibre

Answer: c) Plant fibre

25. Wool is obtained from

- a) Plants
- b) Insects
- c) Animals
- d) Minerals

Answer: c) Animals

26. Wool keeps us warm because

- a) It is heavy
- b) It traps air
- c) It absorbs heat

d) It is shiny

Answer: b) It traps air

27. Silk is obtained from

a) Sheep

b) Cotton plant

c) Silkworm

d) Spider

Answer: c) Silkworm

28. Rearing of silkworms is called

a) Apiculture

b) Sericulture

c) Pisciculture

d) Floriculture

Answer: b) Sericulture

29. The cultivated silk variety is

a) Taser

b) Eri

c) Mulberry silk

d) Muga

Answer: c) Mulberry silk

30. The killing of larva in cocoon stage helps to

a) Increase colour

b) Prevent breaking of silk filament

c) Strengthen fibre

d) Improve smell

Answer: b) Prevent breaking of silk filament

31. Reeling refers to

- a) Boiling cocoon
- b) Rolling silk fibre
- c) Cutting silk
- d) Weaving silk

Answer: b) Rolling silk fibre

32. Raw silk contains

- a) Cellulose
- b) Keratin
- c) Sericin
- d) Lignin

Answer: c) Sericin

33. Rayon is prepared from

- a) Petroleum
- b) Coal
- c) Wood pulp
- d) Cotton lint

Answer: c) Wood pulp

34. Rayon is known as

- a) Artificial wool
- b) Artificial silk
- c) Natural fibre
- d) Plastic fibre

Answer: b) Artificial silk

35. Rayon is a

- a) Natural fibre

b) Fully synthetic fibre

c) Semi-synthetic fibre

d) Mineral fibre

Answer: c) Semi-synthetic fibre

36. Nylon was the

a) First natural fibre

b) First synthetic fibre

c) First plant fibre

d) First mineral fibre

Answer: b) First synthetic fibre

37. Nylon is a

a) Polyamide

b) Polyester

c) Polyethylene

d) Polystyrene

Answer: a) Polyamide

38. Nylon is widely used because it is

a) Weak

b) Brittle

c) Strong

d) Soluble

Answer: c) Strong

39. Polyester is made from

a) Ester units

b) Amide units

c) Glucose units

d) Protein units

Answer: a) Ester units

40. PET stands for

a) Polyethylene Terephthalate

b) Polyethyl Transfer

c) Polymer Ethylene

d) Plastic Ethane

Answer: a) Polyethylene Terephthalate

41. Acrylic fibres are also called

a) Artificial silk

b) Artificial wool

c) Semi-synthetic fibre

d) Cellulose fibre

Answer: b) Artificial wool

42. Acrylic fibres are made from

a) Cotton

b) Acrylonitrile

c) Cellulose

d) Protein

Answer: b) Acrylonitrile

43. Synthetic fibres are generally

a) Biodegradable

b) Non-durable

c) Durable

d) Weak

Answer: c) Durable

44. Synthetic fibres melt before

- a) Stretching
- b) Burning
- c) Wrinkling
- d) Shrinking

Answer: b) Burning

45. Plastics are

- a) Metals
- b) Ceramics
- c) Polymeric materials
- d) Minerals

Answer: c) Polymeric materials

46. Plastics are poor conductors of

- a) Light
- b) Sound
- c) Heat and electricity
- d) Air

Answer: c) Heat and electricity

47. Plastics are widely used because they are

- a) Heavy
- b) Corrosive
- c) Lightweight
- d) Reactive

Answer: c) Lightweight

48. Thermoplastics can be

- a) Reused by heating

- b) Burnt easily
- c) Broken permanently
- d) Dissolved in water

Answer: a) Reused by heating

49. PVC is a

- a) Thermoset
- b) Natural polymer
- c) Thermoplastic
- d) Metal

Answer: c) Thermoplastic

50. Thermosets cannot be

- a) Used
- b) Heated
- c) Recycled
- d) Stored

Answer: c) Recycled

51. Bakelite is an example of

- a) Thermoplastic
- b) Natural polymer
- c) Thermoset
- d) Fibre

Answer: c) Thermoset

52. Resin code number 1 represents

- a) PVC
- b) PET
- c) PS

d) PP

Answer: b) PET

53. HDPE corresponds to resin code

a) 1

b) 2

c) 3

d) 4

Answer: b) 2

54. PVC is represented by resin code

a) 2

b) 3

c) 4

d) 5

Answer: b) 3

55. LDPE is resin code number

a) 4

b) 5

c) 6

d) 7

Answer: a) 4

56. Polypropylene belongs to resin code

a) 5

b) 6

c) 7

d) 1

Answer: a) 5

57. Polystyrene is

- a) Biodegradable
- b) Single-use plastic
- c) Natural polymer
- d) Fibre

Answer: b) Single-use plastic

58. Microplastics are

- a) Large plastics
- b) Broken plastic particles
- c) Natural fibres
- d) Metals

Answer: b) Broken plastic particles

59. PLA is obtained from

- a) Petroleum
- b) Metals
- c) Plant starch
- d) Coal

Answer: c) Plant starch

60. PLA is

- a) Non-biodegradable
- b) Compostable
- c) Thermoset
- d) Metal

Answer: b) Compostable

61. Plastic eating bacteria discovered in 2016 is

- a) E.coli

b) Ideonella sakaiensis

c) Lactobacillus

d) Bacillus

Answer: b) Ideonella sakaiensis

62. PETase enzyme breaks PET into

a) Carbon dioxide

b) Monomers

c) Polymers

d) Ash

Answer: b) Monomers

63. Glass is an

a) Organic material

b) Metal

c) Inorganic solid

d) Fibre

Answer: c) Inorganic solid

64. Glass is

a) Crystalline

b) Amorphous

c) Liquid

d) Elastic

Answer: b) Amorphous

65. Glass can be recycled

a) Once

b) Twice

c) Many times

d) Never

Answer: c) Many times

66. Hard glass is also called

a) Soda glass

b) Lead glass

c) Borosilicate glass

d) Fibre glass

Answer: c) Borosilicate glass

67. Pyrex is an example of

a) Soft glass

b) Hard glass

c) Plastic

d) Fibre

Answer: b) Hard glass

68. Soft glass is also known as

a) Borosilicate

b) Lead glass

c) Soda-lime glass

d) Quartz

Answer: c) Soda-lime glass

69. Window glass is usually

a) Hard glass

b) Soft glass

c) Plastic

d) Fibre

Answer: b) Soft glass

70. Limestone is added to glass to

- a) Increase melting point
- b) Prevent dissolving
- c) Add colour
- d) Increase weight

Answer: b) Prevent dissolving

71. Toughened glass is produced by

- a) Slow cooling
- b) Fast cooling
- c) Adding plastic
- d) Adding salt

Answer: b) Fast cooling

72. Plastic disposal principle includes

- a) 3R
- b) 4R
- c) 5R
- d) 6R

Answer: c) 5R

73. Refuse means

- a) Use plastic
- b) Avoid plastic
- c) Burn plastic
- d) Store plastic

Answer: b) Avoid plastic

74. Recycling plastic helps to

- a) Increase waste

b) Reduce pollution

c) Increase landfills

d) Produce toxins

Answer: b) Reduce pollution

75. Plastic waste in landfills is about

a) 50%

b) 79%

c) 30%

d) 10%

Answer: b) 79%

76. Incineration of plastic causes

a) Soil fertility

b) Air pollution

c) Water purification

d) Composting

Answer: b) Air pollution

77. Plastics are non-biodegradable because

a) Bacteria decompose them

b) They dissolve in water

c) Bacteria cannot decompose them

d) They evaporate

Answer: c) Bacteria cannot decompose them

78. Compostable plastics break down into

a) Toxins

b) Natural elements

c) Metals

d) Ash

Answer: b) Natural elements

79. Synthetic fibres are hydrophobic means

a) Absorb water

b) Repel water

c) Dissolve in water

d) Attract water

Answer: b) Repel water

80. Polyester fabrics are popular because they are

a) Weak

b) Wrinkle-free

c) Expensive

d) Biodegradable

Answer: b) Wrinkle-free

81. Nylon is affected by

a) Water

b) Sunlight

c) Oil

d) Cold

Answer: b) Sunlight

82. Acrylic clothes are preferred because they are

a) Heavy

b) Cheap

c) Weak

d) Absorbent

Answer: b) Cheap

83. PET bottles are mainly used for

- a) Medicines
- b) Water and beverages
- c) Cement
- d) Paint

Answer: b) Water and beverages

84. Polymers used as fibres are

- a) Brittle
- b) Flexible
- c) Rigid
- d) Crystalline

Answer: b) Flexible

85. Glass is a good

- a) Conductor
- b) Semiconductor
- c) Insulator
- d) Magnet

Answer: c) Insulator

86. Fibre obtained from jute is

- a) Animal fibre
- b) Plant fibre
- c) Synthetic fibre
- d) Mineral fibre

Answer: b) Plant fibre

87. Parachutes are made from

- a) Cotton

b) Wool

c) Nylon

d) Jute

Answer: c) Nylon

88. Microplastics mainly pollute

a) Air only

b) Soil and water

c) Fire

d) Rocks

Answer: b) Soil and water

89. Synthetic fibres are not affected by

a) Moisture

b) Moths

c) Chemicals

d) All of these

Answer: d) All of these

90. Plastic production releases

a) Oxygen

b) Nitrogen

c) Greenhouse gases

d) Water vapour

Answer: c) Greenhouse gases

91. Glass is transparent because it

a) Absorbs light

b) Reflects light

c) Allows light to pass

d) Emits light

Answer: c) Allows light to pass

92. PETase enzyme breaks plastic into

a) Polymers

b) Fibres

c) Monomers

d) Metals

Answer: c) Monomers

93. Plastic pollution affects

a) Only oceans

b) Only soil

c) Food chain

d) Rocks

Answer: c) Food chain

94. Nylon melts at

a) Low temperature

b) High temperature

c) Freezing point

d) Room temperature

Answer: b) High temperature

95. Rayon weakens when

a) Dry

b) Wet

c) Heated

d) Frozen

Answer: b) Wet

96. Glass used in lab apparatus is

- a) Soft glass
- b) Lead glass
- c) Borosilicate glass
- d) Plastic

Answer: c) Borosilicate glass

97. Fibres used to make carpets include

- a) Cotton
- b) Wool
- c) Nylon
- d) All of these

Answer: d) All of these

98. Plastics are mouldable due to

- a) Rigidity
- b) Elasticity
- c) Heat and pressure
- d) Transparency

Answer: c) Heat and pressure

99. Plastic bags mainly cause

- a) Soil fertility
- b) Water absorption
- c) Pollution
- d) Composting

Answer: c) Pollution

100. Best solution to plastic pollution is

- a) Burning plastics

- b) Dumping plastics
- c) Following 5R principle
- d) Ignoring plastics

Answer: c) Following 5R principle